

OSMOSIS

PROFICIENT

PHY101

MCQS

PHY 101 – 100 LEVEL

**TOPICS: UNITS & DIMENSIONS,
SCALARS & VECTORS,**

**KINEMATICS, DYNAMICS,
GRAVITATION.**

1. If velocity v has dimensions $[L T^{-1}]$ and time t has $[T]$, what is the dimensional formula of $v^2 t^2$?

A. $[L^2]$

B. $[L^2 T^{-2}]$

C. $[L T^{-1}]$

D. $[L T]$

2. Two vectors $A = 3i + 4j + k$, $B = i - 2j + 2k$. The angle between them is:

A. $\cos^{-1}(0.5)$

B. $\cos^{-1}(1/3)$

C. $\cos^{-1}(1/\sqrt{3})$

D. $\cos^{-1}(2/3)$

3. A particle moves along a path defined by $\mathbf{r}(t) = (t^2)\mathbf{i} + (t^3)\mathbf{j}$. Its velocity at $t = 2$ s is:

A. $\sqrt{4 + 12}$ m/s

B. $\sqrt{16 + 64}$ m/s

C. $\sqrt{4 + 9}$ m/s

D. $\sqrt{2 + 3}$ m/s

4. A body of mass 5 kg is subjected to two forces $F_1 = 3\mathbf{i} + 4\mathbf{j}$ N and $F_2 = -\mathbf{i} + 2\mathbf{j}$ N. Its acceleration is:

A. $(2\mathbf{i} + 6\mathbf{j})/5 \text{ m/s}^2$

B. $(4\mathbf{i} + 2\mathbf{j})/5 \text{ m/s}^2$

C. $(2\mathbf{i} + 3\mathbf{j})/5 \text{ m/s}^2$

D. $(5\mathbf{i} + 6\mathbf{j})/5 \text{ m/s}^2$

5. If $s = 2t^3 - 3t^2 + 4t$, the maximum velocity occurs at:

A. $t = 1 \text{ s}$

B. $t = 2 \text{ s}$

C. $t = 0 \text{ s}$

D. $t = 3 \text{ s}$

6. A satellite orbits Earth at a height h where the acceleration due to gravity is half of g . The orbital period T is proportional to:

A. $(R + h)^{(3/2)}$

B. $(R + h)^{(1/2)}$

C. $(R + h)^2$

D. $(R + h)$

7. A particle moves under acceleration $a = 6t \mathbf{i} + 4 \mathbf{j}$. Its displacement after $t = 2 \text{ s}$, starting from rest, is:

A. $(12\mathbf{i} + 8\mathbf{j}) \text{ m}$

B. $(24\mathbf{i} + 8\mathbf{j}) \text{ m}$

C. $(12\mathbf{i} + 16\mathbf{j}) \text{ m}$

D. $(24\mathbf{i} + 16\mathbf{j}) \text{ m}$

8. A projectile is fired at an angle θ with speed v . The maximum height is H . If g is

halved, keeping v and θ same,
the new height is:

A. $2H$

B. $H/2$

C. H

D. $4H$

9. The dimensional formula of
gravitational potential energy
per unit mass is:

A. $[L^2 T^{-2}]$

B. $[M L^2 T^{-2}]$

C. $[L T^{-1}]$

D. $[M L T^{-2}]$

10. A body of mass m slides down a frictionless inclined plane of angle θ . The component of weight along the incline is:

A. $mg \cos \theta$

B. $mg \sin \theta$

C. $mg \tan \theta$

D. $mg \sec \theta$

11. A particle moves along x-axis with $x = 3t^2 - 2t^3$. The time at which acceleration is zero:

A. 1 s

B. 0 s

C. 2 s

D. 3 s

12. Two forces 10 N and 15 N act at a point with an angle of 120° between them. The resultant is:

A. $5\sqrt{3}$ N

B. 20 N

C. $15\sqrt{3}$ N

D. 25 N

13. If the orbital radius of a satellite is doubled, its orbital speed changes by factor:

A. $1/\sqrt{2}$

B. $\sqrt{2}$

C. $1/2$

D. 2

14. A particle moves with velocity $\mathbf{v} = (3t^2)\mathbf{i} + (4t)\mathbf{j}$. Its speed at $t = 2$ s is:

A. $\sqrt{36 + 16}$ m/s

B. $\sqrt{12 + 8}$ m/s

C. $\sqrt{36 + 64}$ m/s

D. $\sqrt{12 + 16}$ m/s

15. The work done by a force $\mathbf{F} = 10\mathbf{i} + 20\mathbf{j}$ N along displacement $\mathbf{d} = 2\mathbf{i} + \mathbf{j}$ m is:

A. 50 J

B. 40 J

C. 30 J

D. 60 J

16. A particle moves such that its velocity $v = 5 - 2t$. The distance covered before it stops is:

A. 3.125 m

B. 6.25 m

C. 12.5 m

D. 5 m

17. A body moves in a circle of radius r with angular acceleration α . The tangential acceleration is:

A. α/r

B. αr

C. αr^2

D. r/α

18. A vector A has magnitude 5 and vector B magnitude 12. If the resultant is 13, the angle between them is:

A. 90°

B. 60°

C. 30°

D. 45°

19. Two planets have masses M_1 and M_2 and radii R_1 and R_2 . If surface gravities are equal, the ratio of masses is:

A. $M_1/M_2 = (R_1/R_2)^2$

B. $M_1/M_2 = (R_1/R_2)^3$

C. $M_1/M_2 = R_1/R_2$

D. $M_1/M_2 = \sqrt{R_1/R_2}$

20. A particle moves such that $\mathbf{r}(t) = (t^2)\mathbf{i} + (t^3)\mathbf{j}$. Its acceleration vector at $t = 1$ s is:

A. $(2\mathbf{i} + 6\mathbf{j})$

B. $(1\mathbf{i} + 3\mathbf{j})$

C. $(4\mathbf{i} + 9\mathbf{j})$

D. $(2\mathbf{i} + 3\mathbf{j})$

21. The period of a simple pendulum increases by 10% due to increase in length. The

**percentage increase in length
is approximately:**

A. 21%

B. 10%

C. 5%

D. 15%

**22. A particle moves under
force $F = k r$. The motion is:**

A. Uniform circular

B. Simple harmonic

C. Uniform linear

D. Parabolic

23. The velocity of a satellite in elliptical orbit is maximum at:

A. Apogee

B. Perigee

C. Center

D. Equally at both ends

24. A particle moves under acceleration $a = 4t \mathbf{i} + 6 \mathbf{j}$. The velocity after 3 s, starting from rest, is:

A. 36 m/s

B. 30 m/s

C. 24 m/s

D. 42 m/s

25. A projectile is launched horizontally from height h with speed v . Time of flight is:

A. $\sqrt{2h/g}$

B. v/g

C. $\sqrt{h/g}$

D. $\sqrt{4h/g}$

26. A particle moves along x-axis: $x = 4t^3 - 3t^2 + 2t$.

Acceleration at $t = 1$ s:

A. 18 m/s^2

B. 12 m/s^2

C. 24 m/s^2

D. 10 m/s^2

27. Two vectors $A = i + j + k$, $B = 2i - j + k$. Their cross product magnitude is:

A. $\sqrt{6}$

B. $\sqrt{2}$

C. $\sqrt{3}$

D. 2

28. If the radius of Earth is doubled, keeping mass same, acceleration due to gravity changes by factor:

A. $1/4$

B. $1/2$

C. 2

D. 4

29. A particle is subjected to force $F = 10 \text{ N}$ along displacement of 3 m at angle 60° . Work done is:

A. 15 J

B. 10 J

C. 5 J

D. 20 J

30. A particle of mass m moves under force $F = m(g + a)$. Its weight on an accelerating elevator is:

A. mg

B. $m(g + a)$

C. $m(g - a)$

D. Zero

31. A particle moves along curve $y = x^2$. At $x = 2$ m, the slope of the tangent is:

A. 4

B. 2

C. 1

D. 0

32. Two bodies of masses m and $2m$ collide elastically. The fraction of kinetic energy transferred to $2m$ is:

A. $4/9$

B. $1/3$

C. $2/3$

D. $1/2$

33. The gravitational potential energy of a satellite of mass m at distance r from Earth's center:

- A. $-GMm/r$**
- B. GMm/r^2**
- C. GMm/r**
- D. $-GMm/r^2$**

34. A particle moves along x-axis with $x = 3t^2 - 2t^3$. At what time is acceleration zero?

- A. 1 s**
- B. 2 s**
- C. 3 s**
- D. 0 s**

35. A particle moves under acceleration $a = 6t$. Initial velocity zero. Displacement after $t = 2$ s is:

A. 4 m

B. 8 m

C. 12 m

D. 16 m

36. The component of vector $A = 2i + 3j$ along vector $B = i + j$ is:

A. $5/\sqrt{2}$

B. $\sqrt{13}$

C. 3

D. 2

37. A particle is dropped from height h . Velocity after t seconds:

A. gt

B. $\sqrt{2gh}$

C. $2gt$

D. gt^2

38. The centripetal acceleration of a particle moving in circle of radius 4 m at 2 m/s is:

A. 1 m/s^2

B. 2 m/s^2

C. 1 m/s^2

D. 4 m/s^2

39. Two vectors A and B of equal magnitude form an angle θ . The magnitude of

resultant is $\sqrt{3}$ times one vector. θ is:

A. 60°

B. 30°

C. 45°

D. 90°

40. The period of a simple pendulum is $T = 2$ s. Its length is:

A. 1 m

B. 0.99 m

C. 1.5 m

D. 2 m

41. The escape velocity from a planet of radius R and surface gravity g is:

A. $\sqrt{2gR}$

B. $\sqrt{gR}$

C. $2gR$

D. gR

42. A particle moves with velocity $v = t^2 - 4t + 5$.

Acceleration at $t = 3$ s is:

A. 2 m/s^2

B. 1 m/s^2

C. 0

D. -2 m/s^2

43. A particle moves along circle of radius $r = 5 \text{ m}$ with angular speed $\omega = 2 \text{ rad/s}$. Centripetal acceleration is:

A. 10 m/s^2

B. 20 m/s^2

C. 5 m/s^2

D. 15 m/s^2

44. The resultant of vectors $2\mathbf{i} + 3\mathbf{j}$ and $4\mathbf{i} - \mathbf{j}$ is:

A. $6\mathbf{i} + 2\mathbf{j}$

B. $2\mathbf{i} + 4\mathbf{j}$

C. $6\mathbf{i} + 4\mathbf{j}$

D. $8\mathbf{i} + 2\mathbf{j}$

45. A particle moves under acceleration $a = 3t^2$.

Displacement after $t = 2$ s, starting from rest, is:

A. 4 m

B. 8 m

C. 6 m

D. 12 m

46. The gravitational force between two masses $m_1 = 3$ kg and $m_2 = 6$ kg separated by 2 m is:

A. 1.5×10^{-10} N

B. 6×10^{-11} N

C. 3×10^{-11} N

D. 2×10^{-10} N

47. Two projectiles are fired with same speed v at angles θ and $90^\circ - \theta$. Their ranges R_1 and R_2 are related as:

- A. $R_1 = R_2$**
- B. $R_1 > R_2$**
- C. $R_1 < R_2$**
- D. No relation**

48. A particle moves along $y = x^3$. Slope of tangent at $x = 2$ is:

- A. 12**
- B. 8**

C. 6

D. 4

49. The magnitude of vector $(i + j) \times (2i - j)$ is:

A. 3

B. 1

C. 2

D. $\sqrt{3}$

50. A body moves along x-axis with acceleration $a = 5t$.

Initial velocity zero.

Displacement after 3 s is:

A. 22.5 m

B. 15 m

C. 20 m

D. 30 m